# Customer Shopping Behaviour Analysis

## 1. Project Overview

This project analyzes customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

## 2. Dataset Summary

- Rows: 3,900 - Columns: 18 - Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

## 3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.

- **Initial Exploration:** Used `df.info()` to check structure and `.describe(include="all")` for summary statistics.

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
 4   Category                3900 non-null   str
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   str
 7   Size                    3900 non-null   str
 8   Color                   3900 non-null   str
 9   Season                  3900 non-null   str
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str
 12  Shipping Type           3900 non-null   str
 13  Discount Applied        3900 non-null   str
 14  Promo Code Used         3900 non-null   str
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   str
 17  Frequency of Purchases  3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

| | Customer ID | Age | Gender | Item Purchased | Category | Purchase Amount (USD) | Location | Size | Color | Season | Review Rating | Subscription Status | Shippi Ty |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| count | 3900.000000 | 3900.000000 | 3900 | 3900 | 3900 | 3900.000000 | 3900 | 3900 | 3900 | 3900 | 3863.000000 | 3900 | 39 |
| unique | NaN | NaN | 2 | 25 | 4 | NaN | 50 | 4 | 25 | 4 | NaN | 2 | |
| top | NaN | NaN | Male | Blouse | Clothing | NaN | Montana | M | Olive | Spring | NaN | No | Fr Shippi |
| freq | NaN | NaN | 2652 | 171 | 1737 | NaN | 96 | 1755 | 177 | 999 | NaN | 2847 | 6 |
| mean | 1950.500000 | 44.068462 | NaN | NaN | NaN | 59.764359 | NaN | NaN | NaN | NaN | 3.750065 | NaN | N |
| std | 1125.977353 | 15.207589 | NaN | NaN | NaN | 23.685392 | NaN | NaN | NaN | NaN | 0.716983 | NaN | N |
| min | 1.000000 | 18.000000 | NaN | NaN | NaN | 20.000000 | NaN | NaN | NaN | NaN | 2.500000 | NaN | N |
| 25% | 975.750000 | 31.000000 | NaN | NaN | NaN | 39.000000 | NaN | NaN | NaN | NaN | 3.100000 | NaN | N |
| 50% | 1950.500000 | 44.000000 | NaN | NaN | NaN | 60.000000 | NaN | NaN | NaN | NaN | 3.800000 | NaN | N |
| 75% | 2925.250000 | 57.000000 | NaN | NaN | NaN | 81.000000 | NaN | NaN | NaN | NaN | 4.400000 | NaN | N |
| max | 3900.000000 | 70.000000 | NaN | NaN | NaN | 100.000000 | NaN | NaN | NaN | NaN | 5.000000 | NaN | N |

| Shipping Type | Discount Applied | Promo Code Used | Previous Purchases | Payment Method | Frequency of Purchases |
|---|---|---|---|---|---|
| 3900 | 3900 | 3900 | 3900.000000 | 3900 | 3900 |
| 6 | 2 | 2 | NaN | 6 | 7 |
| Free Shipping | No | No | NaN | PayPal | Every 3 Months |
| 675 | 2223 | 2223 | NaN | 677 | 584 |
| NaN | NaN | NaN | 25.351538 | NaN | NaN |
| NaN | NaN | NaN | 14.447125 | NaN | NaN |
| NaN | NaN | NaN | 1.000000 | NaN | NaN |
| NaN | NaN | NaN | 13.000000 | NaN | NaN |
| NaN | NaN | NaN | 25.000000 | NaN | NaN |
| NaN | NaN | NaN | 38.000000 | NaN | NaN |
| NaN | NaN | NaN | 50.000000 | NaN | NaN |

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Feature Engineering:**

- Created **age_group** column by binning customer ages.
- Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.

- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

## 4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

Result Grid | Fil

| gender | revenue |
|---|---|
| Male | 157890 |
| Female | 75191 |

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

Result Grid | Filter Rows:

| customer_id | purchase_amount |
|---|---|
| 2 | 64 |
| 3 | 73 |
| 4 | 90 |
| 7 | 85 |
| 9 | 97 |
| 12 | 68 |
| 13 | 72 |
| 16 | 81 |
| 20 | 90 |
| 22 | 62 |
| 24 | 88 |
| 29 | 94 |

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

| item_purchased | Average Product Rating |
|---|---|
| Gloves | 3.86 |
| Sandals | 3.84 |
| Boots | 3.82 |
| Hat | 3.8 |
| Skirt | 3.78 |

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

| shipping_type | ROUND(AVG(purchase_amount),2) |
|---|---|
| Express | 60.48 |
| Standard | 58.46 |

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

| subscription_status | total_customers | avg_spend | total_revenue |
|---|---|---|---|
| Yes | 1053 | 59.49 | 62645 |
| No | 2847 | 59.87 | 170436 |

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

Result Grid  Filter Rows:

| item_purchased | discount_rate |
|---|---|
| Hat | 50.00 |
| Sneakers | 49.66 |
| Coat | 49.07 |
| Sweater | 48.17 |
| Pants | 47.37 |

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

| customer_segment | Number of Customers |
|---|---|
| Loyal | 3116 |
| Returning | 701 |
| New | 83 |

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

| item_rank | category | item_purchased | total_orders |
|---|---|---|---|
| 1 | Accessories | Jewelry | 171 |
| 2 | Accessories | Sunglasses | 161 |
| 3 | Accessories | Belt | 161 |
| 1 | Clothing | Blouse | 171 |
| 2 | Clothing | Pants | 171 |
| 3 | Clothing | Shirt | 169 |
| 1 | Footwear | Sandals | 160 |
| 2 | Footwear | Shoes | 150 |
| 3 | Footwear | Sneakers | 145 |
| 1 | Outerwear | Jacket | 163 |
| 2 | Outerwear | Coat | 161 |

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

| subscription_status | repeat_buyers |
|---|---|
| Yes | 958 |
| No | 2518 |

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

| age_group | total_revenue |
|---|---|
| Young Adult | 62143 |
| Middle-aged | 59197 |
| Adult | 55978 |
| Senior | 55763 |

## 5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

## 6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.